Shahjalal University of Science and Technology, Sylhet

Department of Electrical & Electronic Engineering

Course Code: EEE 0713 3237 Course Title: Power System II

Semester Final Examination: **Model Question**

SET A

Practice paper for self-study, made from previous finals, term tests and lecture notes. Not an official question paper.

Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

[The figures in the right margin indicate full marks. Answer Q1, Q2 and either Q3 or Q4 from Part A; Q5, Q8 and either Q6 or Q7 from Part B.]

Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

1. (a) What is transposition of a transmission line? Why is it used? [2]

(b) Derive the expression for the inductance per phase of a fully transposed three-phase line with unsymmetrical spacing. [4]

(c) Each conductor of the bundled-conductor line shown in Fig. 1 is ACSR, 1,272,000-cmil Pheasant (GMR = 0.0466 ft). Find the inductive reactance in Ω/km and Ω/mile per phase for d = 45 cm, f = 60 Hz. Also find the per-unit series reactance if the line is 160 km long (base 100 MVA, 345 kV). [4]

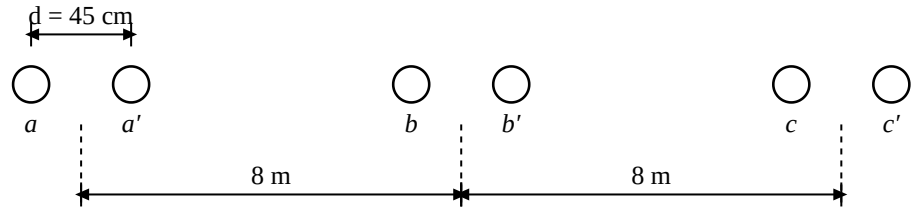


Fig. 1

2. (a) How does the earth affect the capacitance of a transmission line? [2]

(b) Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing. [5]

(c) A three-phase 50 Hz transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductors are No. 2/0 hard-drawn seven-strand copper (outside diameter = 1.05 cm). The line voltage is 110 kV. Find the capacitance to neutral and the charging current per km of line. [3]

3. (a) Write down the general construction of a 3-core cable with a neat figure. [3]

(b) Derive the expression for dielectric stress in a single-core cable and prove that it is maximum at the conductor surface. [4]

(c) A single-core cable for use on 11 kV, 50 Hz has a conductor area of 0.645 cm² and an internal sheath diameter of 2.18 cm. The relative permittivity of the dielectric is 3.5. Find (i) g_{max}, (ii) g_{min}, (iii) capacitance per km and (iv) charging current. [3]

OR

4. (a) Derive the condition for the most economical conductor size in a cable. [3]

(b) Find the most economical size of a single-core cable working on a 132 kV, 3-phase system if a dielectric stress of 60 kV/cm can be allowed. [2]

(c) The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and the resistivity of insulation is 4.5×10^{14} Ω-cm, find the insulation thickness. [3]

(d) Define grading of cables. Why is unequal stress distribution in a cable undesirable? [2]

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

5. (a) What is string efficiency? Discuss the methods of improving string efficiency. [4]

(b) A 3-phase transmission line is supported by three-disc insulators. The potentials across the top unit (near the tower) and the middle unit are 8 kV and 11 kV respectively. (i) Draw the equivalent circuit, (ii) find the ratio of capacitance between pin and earth to the self-capacitance of each unit, (iii) find the line voltage and (iv) the string efficiency. [6]

6. (a) Define critical disruptive voltage. Write down its significance. [2]

(b) What is corona? Write down its advantages and disadvantages. [3]

(c) The towers of height 30 m and 90 m respectively support a transmission line conductor at a water crossing. The horizontal distance between the towers is 500 m. If the tension in the conductor is 1600 kg, find the minimum clearance of the conductor from the water and the clearance mid-way between the supports. The weight of the conductor is 1.5 kg/m. The bases of the towers can be considered to be at water level. [5]

OR

7. (a) Classify the types of cable faults. [2]

(b) Explain the Murray loop test with a diagram and derive the expression for the distance of the fault. [3]

(c) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistances of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end. [3]

(d) A Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 Ω. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 Ω and the fixed resistors are equal. Find the distance of the fault. [2]

8. (a) Derive the swing equation for a synchronous machine. [4]

(b) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus (|V| = 1.0 pu) with terminal voltage $|V_t|$ = 1.0 pu. Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred when (i) the system is healthy, (ii) one line is shorted (3-phase) in the middle and (iii) one line is open. [6]

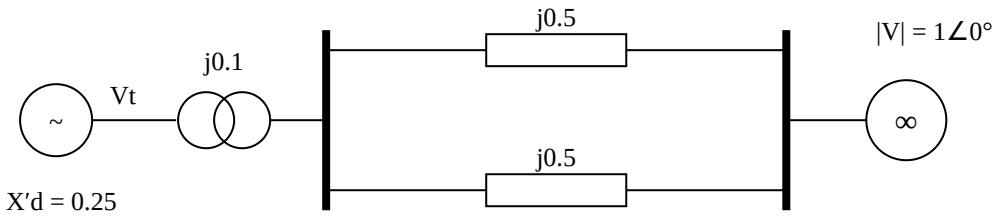


Fig. 2

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SET B

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Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

[The figures in the right margin indicate full marks. Answer Q1, Q2 and either Q3 or Q4 from Part A; Q5, Q8 and either Q6 or Q7 from Part B.]

Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

1. (a) Define inductance with the required expression. What is skin effect? [2]

(b) Derive the expression for the inductance of composite-conductor lines. [5]

(c) A single-phase 50 Hz power line is supported on a horizontal cross-arm with 2.5 m spacing between conductors. A telephone line (solid copper conductors, 0.6 m apart) is on a cross-arm in the same horizontal plane. The distance between the nearest conductors of the two lines is 20 m (Fig. 1). Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150 A in the power line. [3]

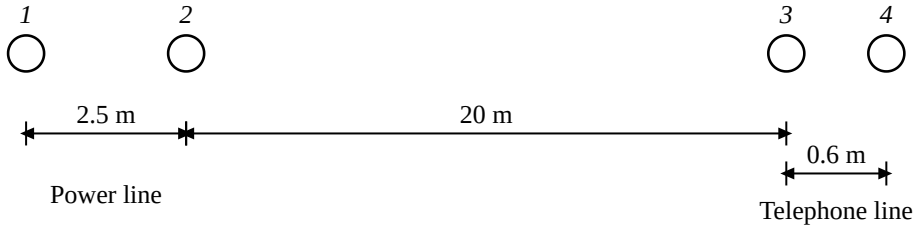


Fig. 1

2. (a) What is a bundled conductor? Why is it used? [2]

(b) A 500 kV, 50 Hz line has a bundling arrangement of 2 conductors per phase as shown in Fig. 2. Compute the inductive reactance per phase. Each conductor carries 50 % of the phase current. [3]

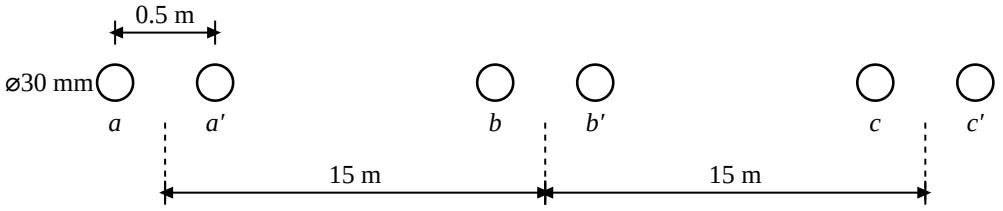


Fig. 2

(c) Derive the expression for the capacitance of a three-phase line with equilateral spacing. [3]

(d) Find the capacitive susceptance per mile of a single-phase line operating at 60 Hz. The conductor is Partridge (outside diameter 0.642 in) and the spacing is 20 ft between centres. [2]

3. (a) Write down the advantages and the necessary requirements of underground cables. [2]

(b) Explain capacitance grading. Considering three dielectric layers, derive the expression for the potential difference between core and earthed sheath. [5]

(c) A 66 kV single-core lead-sheathed cable is graded by using two dielectrics of relative permittivity 5 and 3, the thickness of each being 1 cm. The core diameter is 2 cm. Determine the maximum stress in the two dielectrics. [3]

OR

4. (a) Derive the expression for the insulation resistance of a single-core cable. [3]

(b) Determine the insulation resistance of a single-core cable of length 3 km having conductor radius 12.5 mm, insulation thickness 10 mm and specific resistance of insulation 5×10^{12} Ω-m. [2]

(c) The capacitances of a 3-phase belted cable are 12.6 μF between the three cores bunched together and the lead sheath, and 7.4 μF between one core and the other two connected to the sheath. Find the charging current drawn by the cable when connected to a 66 kV, 50 Hz supply. [3]

(d) A 33 kV single-core cable has a conductor diameter of 1 cm and a sheath of inside diameter 4 cm. Find the maximum and minimum stress in the insulation. [2]

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

5. (a) Name the types of insulators. Why are suspension insulators used above 33 kV? [2]

(b) Derive the mathematical expression of string efficiency with a proper diagram. What will happen if (i) K = 0.2, (ii) K = 0.8, (iii) the number of discs in the string increases? (K = shunt capacitance / self capacitance) [5]

(c) In a 33 kV overhead line there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11 % of the self-capacitance of each insulator, find (i) the distribution of voltage over the 3 insulators and (ii) the string efficiency. [3]

6. (a) Define sag. What is a stringing chart? [2]

(b) Derive the expressions for sag when the supports are at unequal levels. [3]

(c) A 132 kV transmission line has: weight of conductor = 680 kg/km, span = 260 m, ultimate strength = 3100 kg, safety factor = 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 m. [2]

(d) A transmission line has a span of 300 m between level supports. The conductor has a cross-sectional area of 1.3 cm^2, weighs 1170 kg/km and has a breaking stress of 4218 kg/cm^2. Calculate the sag for a safety factor of 5, allowing a wind pressure of 122 kg per m^2 of projected area. What is the vertical sag? [3]

OR

7. (a) Define thermal resistance of a cable. On what factors does the current-carrying capacity of a cable depend? [2]

(b) A single-core cable is laid in the ground; core diameter 30 mm, dielectric thickness 40 mm. The thermal resistivity of the dielectric is 5 thermal Ω-m and the thermal resistance between sheath and ground surface is 0.45 thermal Ω per m. Neglecting dielectric and sheath losses, estimate the maximum permissible current if the temperature difference between conductor and ground surface must not exceed 55 °C. The electrical resistance of the cable is 110 μΩ per metre. [4]

(c) A Varley loop test is performed to locate an earth fault on a 40 km long cable. The resistance per km of the single conductor is 40 Ω. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 Ω and the fixed resistors are equal. Find the distance of the fault from the test end. [4]

8. (a) Define power angle. Briefly explain the equal-area criterion. [3]

(b) Using the P–δ curve, explain what happens for a sudden increase in mechanical input to a generator. [4]

(c) In Fig. 3 the machine delivers 1.0 pu power and both the terminal voltage and the infinite-bus voltage are 1.0 pu. Determine the power-angle equation for the given operating condition. [3]

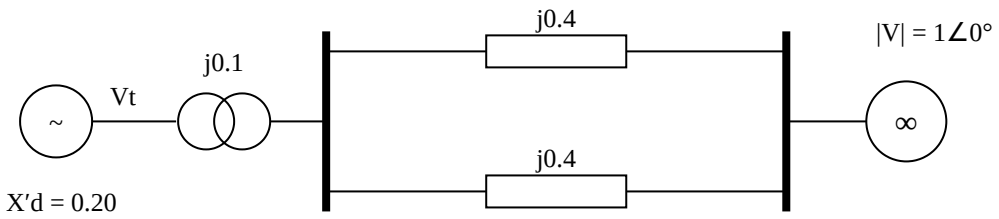


Fig. 3

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Semester Final Examination: **Model Question**

SET C

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Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

[The figures in the right margin indicate full marks. Answer Q1, Q2 and either Q3 or Q4 from Part A; Q5, Q8 and either Q6 or Q7 from Part B.]

Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

1. (a) Between aluminium and copper, which one is superior as an overhead conductor? Explain why. [2]

(b) Derive the expression for the flux linkages of one conductor in a group of conductors. [4]

(c) A double-circuit three-phase line is shown in Fig. 1. Conductors a, a′; b, b′; c, c′ belong to the same phase respectively. The radius of each conductor is 1.9 cm. Find the inductance of the line in mH/km/phase (assume transposition). [4]

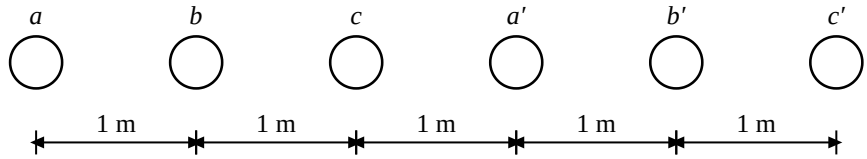


Fig. 1

2. (a) Prove that the inductance of a single-phase two-wire line is $L = 4 \times 10^{-7} \ln(D/r')$ H/m. [4]

(b) A single-phase 50 Hz power line has a spacing of 3 m between conductors. A telephone line is supported symmetrically 2 m below the power line as shown in Fig. 2 (telephone conductor spacing 0.8 m). Find the mutual inductance between the two circuits and the voltage induced per km in the telephone line if the current in the power line is 100 A. [4]

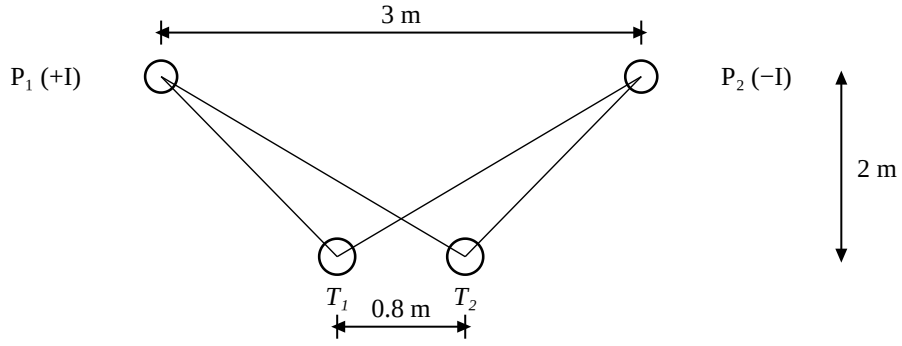


Fig. 2

(c) What is the difference between a composite conductor and a bundled conductor? [2]

3. (a) Define transposition. Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing. [5]

(b) Explain the method of images and show how the presence of earth modifies the line capacitance. [3]

(c) Show that the capacitance to neutral of a single-phase line is twice the line-to-line capacitance. [2]

OR

4. (a) Prove that $g_{max}/g_{min} = D/d$ for a single-core cable, where the symbols have their usual meaning. [2]

(b) The maximum and minimum stresses in the dielectric of a single-core cable are 40 kV/cm (rms) and 10 kV/cm (rms) respectively. If the conductor diameter is 2 cm, find (i) the thickness of insulation and (ii) the operating voltage. [3]

(c) A 33 kV, 50 Hz, 3-phase underground cable, 4 km long, uses three single-core cables. Each conductor has a diameter of 2.5 cm and the radial thickness of insulation is 0.5 cm ($\varepsilon_r = 3$). Determine (i) the capacitance per phase, (ii) the charging current per phase and (iii) the total charging kVAR. [3]

(d) How are C_e and C_c of a 3-core cable measured? [2]

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

5. (a) What are the main components of an overhead line? What kind of supports are used for distribution (up to 11 kV)? [2]

(b) Each line of a 3-phase system is suspended by a string of 3 similar insulators. If the voltage across the line unit is 17.5 kV, calculate the line-to-neutral voltage and the line voltage. Assume the shunt capacitance between each insulator and earth is 1/8 of the capacitance of the insulator itself. Also find the string efficiency. [4]

(c) A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, calculate (i) the distribution of voltage on the insulator discs and (ii) the string efficiency. [4]

6. (a) What is corona? Explain the theory of corona formation and the methods of reducing corona. [3]

(b) Consider a 132 kV line with 1.956 cm diameter conductors, built so that corona takes place if the line voltage exceeds 210 kV (rms). If the potential gradient at which ionisation occurs is 30 kV/cm (peak), find the spacing between the conductors. [3]

(c) An overhead line at a river crossing is supported from two towers of heights 40 m and 90 m above water level, the horizontal distance between the towers being 400 m. If the maximum allowable tension is 2000 kg, find the clearance between the conductor and the water at a point mid-way between the towers. Weight of conductor is 1 kg/m. [4]

OR

7. (a) What are the causes of insulation failure? How does electrical breakdown happen in a pin-type insulator? [3]

(b) Write down the desirable properties of insulators. [2]

(c) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistances of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end, with a neat diagram of the test. [5]

8. (a) Define HVDC. Discuss the economics of power transmission with a cost–distance curve. [3]

(b) Describe different FACTS devices and their applications. [2]

(c) The generator of Fig. 3 is delivering 1.0 pu power to the infinite bus (|V| = 1.0 pu) with terminal voltage $|V_t|$ = 1.0 pu. Calculate the emf behind transient reactance and find the maximum power that can be transferred when (i) the system is healthy, (ii) one line is shorted (3-phase) in the middle and (iii) one line is open. Plot all three power-angle curves. [5]

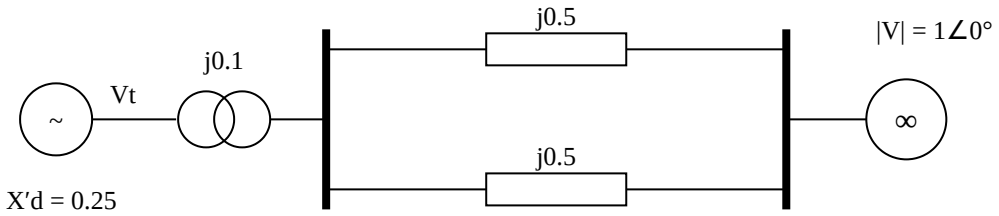


Fig. 3

Answer Key (final numerical answers)

Full step-by-step solutions: **03_Solved_Math_Bank.md**. Derivation steps: **04_Derivations_Guide.md**. Theory answers: **05_Short_Questions_and_Definitions.md**.

Set A

- 1(c) X = 0.365 Ω/km = 0.587 Ω/mile; X_{pu} = 0.049
- 2(c) C_n = 0.00826 µF/km; I_c = 0.165 A/km
- 3(c) g_{max} = 27.66 kV/cm, g_{min} = 11.5 kV/cm, C = 0.222 µF/km, I_c = 0.766 A
- 4(b) d = 3.59 cm, D = 9.77 cm · 4(c) thickness ≈ 1.25 cm
- 5(b) K = 0.375; V_3 = 18.125 kV; V_L = 64.3 kV; η = 68.3 %
- 6(c) min clearance = 23.02 m; mid-way clearance = 30.70 m
- 7(c) 433.3 m · 7(d) 15 km
- 8(b) $E' = 1.075\angle 33.9°$ pu; P_{max} = 1.79 / 0.693 / 1.264 pu

Set B

- 1(c) M = 0.648 µH/km; V = 30.6 mV/km
- 2(b) X = 0.346 Ω/km · 2(d) B = 2.55 µS/mile
- 3(c) g_{1max} = 48.2 kV/cm, g_{2max} = 40.2 kV/cm
- 4(b) 155.9 MΩ · 4(c) C_e = 4.2 µF, C_c = 1.6 µF, I_c = 107.7 A · 4(d) 47.6 / 11.9 kV/cm
- 5(c) V_1, V_2, V_3 = 5.52, 6.13, 7.41 kV; η = 85.7 %
- 6(c) 13.71 m · 6(d) sag = 20.08 m, vertical sag = 12.0 m
- 7(b) I ≈ 580 A · 7(c) 37.5 km
- 8(c) $E' = 1.05\angle 28.4°$; $P_e = 2.10 \sin \delta$ pu

Set C

- 1(c) L = 0.431 mH/km/phase (D_s = 0.2107 m, D_{eq} = 1.815 m)
- 2(b) M = 0.0758 mH/km; V = 2.38 V/km
- 3 Theory/derivation: $C_n = 2\pi k / \ln(D_{eq}/r)$
- 4(b) thickness 3 cm, V = 55.45 kV · 4(c) 1.98 µF, 11.87 A, 678 kVAR
- 5(b) V_{ph} = 44.24 kV, V_L = 76.6 kV, η = 84.3 % · 5(c) 8.00, 8.80, 10.48, 13.21, 17.25 kV; η = 66.9 %
- 6(b) spacing ≈ 3.38 m · 6(c) mid-way clearance = 55 m
- 7(c) 433.3 m
- 8(c) $E' = 1.075\angle 33.9°$ pu; P_{max} = 1.79 / 0.693 / 1.264 pu

Why these 3 sets cover about 70–80 % of the final

Topic (repeat count in past finals + TTs)	Set A	Set B	Set C
Capacitance 3-φ unsymmetrical derivation (5×)	2b	–	3a
Transposition (5×)	1a	–	3a
Inductance unsymmetrical / composite / one-conductor-in-group / two-wire derivations	1b	1b	1b, 2a
Bundled Pheasant (3×) · 500 kV bundle (2×) · double circuit (your TT)	1c	2b	1c
Telephone line, both versions (3×)	–	1c	2b
Flat-spacing capacitance (2×) · earth effect (2×) · Partridge	2a, 2c	2d	3b

Cable stress derivation, g_{max}/g_{min} (3×) · economical size (3×)	3b, 4a–b	4d	4a–b
Insulation resistance / thickness (3×) · grading (2×)	4c–d	3b–c, 4a–b	–
11 kV 0.645 cm² (2×) · belted cable · 33 kV 4 km	3c	4c	4c–d
Murray loop (3×) · Varley loop (3×) · thermal / permissible current	7b–d	7	7c
String efficiency theory (4×) + derivation · string maths (5×)	5	5	5
Sag maths (4×) · sag / stringing chart	6c	6	6c
Corona · critical disruptive voltage (3× each)	6a–b	–	6a–b
Insulators: types, properties, causes of failure, pin-type breakdown	–	5a	7a–b
Swing equation (2×) · P–δ sudden increase (3×) · equal-area criterion	8a	8a–b	–
Generator E′ & P_{max} (3×) · power-angle equation	8b	8c	8c
HVDC economics / types · FACTS	–	–	8a–b

How to use: sit each set as a 2-hour mock. Then check against the answer key and the solution files. Every question here was asked before in a past final or term test, or the teacher flagged it in class. The highest-repeat questions appear in two or three sets on purpose.